

Suryanshu Saini

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering — CGPA 8.78

Vellore, Tamil Nadu

Sep. 2022 – April 2026

WORK EXPERIENCE

Software Engineering Intern

Jan 2026 – June 2026

Accord Technologies - Remote

- Architected a **real-time backend** using **Convex**, leveraging ACID transactions, automatic query invalidation, and live sync to coordinate booking state across concurrent clients without race conditions
- Integrated **Stripe Connect** split-payment flows with deferred capture, handling platform commission splits, direct provider payouts, and idempotent refund and cancellation logic
- Built dual-role service marketplace with real-time booking, live chat, and push notifications for **iOS and Android** using **React Native** and **Expo**

UI/UX Design Intern

May 2024 – July 2024

PhPPoets Pvt. Ltd., Udaipur, Rajasthan

- Delivered 3 responsive landing pages and 7+ custom icon packs for enterprise clients across multiple industries
- Produced 11+ high-fidelity prototypes and 2 end-to-end mobile app designs, iterating directly on client feedback

PROJECTS

WebMellon | Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, Docker, Prometheus, Grafana May 2025 – June 2025

- Built a **full-stack bookmark aggregator and RSS reader** in **Next.js** (App Router) supporting YouTube, WordPress, and major news feeds, with a live deployment at webmellon.vercel.app
- Designed a **REST API layer** using Next.js Route Handlers with **server-side pagination**, search, and multi-field sorting backed by **Prisma ORM** and **PostgreSQL** — reducing initial page load by 70–90%
- Implemented **OAuth 2.0 authentication** via **NextAuth.js** (Google & GitHub providers) with **JWT** session handling and secure cookie middleware
- Containerised the full stack with **Docker Compose** (Next.js + PostgreSQL + Prometheus + Grafana) and set up **GitHub Actions** CI/CD pipelines with automated Vercel deployment
- Wrote end-to-end tests with **Playwright** and unit tests with **Vitest**; integrated **Prometheus** metrics and **Grafana** dashboards for production monitoring

We Go Wayback | Go, Redis, Asynq, Chi, StatsD, Wayback Machine

May 2025 – June 2025

- Ported the Python **wayback-discover-diff** service to **Go**, rewriting the HTTP layer and core diff logic while preserving full API compatibility
- Implemented a **SimHash** module for near-duplicate detection of archived web pages, used to efficiently identify content changes between Wayback Machine snapshots
- Integrated **Asynq** with **Redis** as the task queue backend, using **goroutines** for concurrent job processing; exposed a **REST API** via **go-chi** with **StatsD** instrumentation
- Achieved **100% unit test coverage** across all modules

BCP (Better Context Protocol) | Rust, TypeScript, Python, gRPC, Protocol Buffers

Dec 2025 – Present

- Explored an alternative to JSON-RPC for AI agent tool communication using **gRPC** and **Protocol Buffers**, experimenting with semantic tool discovery via embedding search

TECHNICAL SKILLS

Languages: TypeScript, Go, Python

Frontend: React, Next.js (App Router), SvelteKit, Tailwind CSS, shadcn/ui, Jotai

Backend: Node.js, REST APIs, Next.js Route Handlers, gRPC, WebSockets, Middleware, Hono, Express

Databases & ORM: PostgreSQL, Redis, MongoDB, Prisma

Auth & Security: OAuth 2.0, NextAuth.js, JWT, Secure Cookies

DevOps & Tooling: Docker, Docker Compose, GitHub Actions (CI/CD), Vercel

Testing: Playwright (E2E), Vitest, Unit Testing

Monitoring: Prometheus, Grafana, StatsD